

7 Module Documentation

7.1 Strategic Thinking (M1)

7.1.1 Headline information about the module

Module title	Strategic Thinking
Module NFQ level (only if an NFQ level can be demonstrated)	8
Module number/reference	M1
Parent programme(s) the plural arises if there are embedded programmes to be validated.	Higher Diploma in Science in Data Analytics for Business
Stage of parent programme	AWARD
Semester (semester1/semester2 if applicable)	1
Module credit units (FET/HET/ECTS)	ECTS
Module credit number of units	10
List the teaching and learning modes	Full time & Part time blended learning
Entry requirements (statement of knowledge, skill and competence)	See Section 4
Pre-requisite module titles	none
Co-requisite module titles	none
Is this a capstone module? (Yes or No)	No
Specification of the qualifications (academic, pedagogical and professional/occupational) and experience required of staff (staff includes workplace personnel who are responsible for learners such as apprentices, trainees and learners in clinical placements)	Academic and Professional: MSc required. However, NFQ Level 8 in Computer Science, Software Development, Software Engineering or equivalent is acceptable in cases where significant industrial experience is evident. Industry experience will be essential for those who do not have a postgraduate award. Software Development and Associated Experience: A minimum of 1-year experience required, 3 years of experience preferred. Pedagogical: Teaching experience is desired. Completion of postgraduate CPD/Certificate in Teaching and Learning or similar preferred. Experience in blended learning delivery required. In absence of experience, training will be provided.
Maximum number of learners per centre (or instance of the module)	120

Duration of the module						2 semesters full-time 3 semesters part-time				
Average (over the duration of the module) of the contact hours per week (see * below)						2.5 hours per week over 2 semesters (full-time) 1.7 hours per week over 3 semesters (part-time)				
						Learning will be delivered through a combination of synchronous and asynchronous learning, with 30 hours dedicated to directed e-learning activities (which includes live online classes), 10 hours (minimum) dedicated to face to face and 20 hours of other types of contact hours which can be used flexibly as face to face or online to be determined by the pedagogy, needs of the cohort and the particular activity.				
Module-specific physical resources and support required per centre (or instance of the module)						Physical resource requirements are 1 laptop or PC/workstation per student. On campus and online resources as per programme specification.				
Analysis of required learning effort										
*Effort while in contact with staff										
On campus Lecture / Classroom demonstrations		Mentoring and small-group tutoring		Other (Reflective development, directed reading/group work)		Online classes & Directed e-learning (hours)	Independent learning (hours)	Other hours (specify)	Work-based learning hours of learning effort	Total effort (hours)
Hours	Minimum ratio teacher/learner	Hours	Minimum ratio teacher/learner	Hours	Minimum ratio teacher/learner					
10	1:16	10	1:5	10	1:16	30	190	N/A	N/A	250
Allocation of marks (within the module)										
				Continuous assessment	Supervised project	Proctored practical examination	Proctored written examination	Total		
Percentage contribution				100%				100%		

7.1.2 Module aims and objectives

It is the aim of this module to provide the learner with knowledge of:

1. The macro and micro business environment and the concepts, tools and techniques that facilitate its analysis
2. The relationship between business and information technology and how information technology can establish and maintain a competitive advantage
3. Differences between proprietary and open source software models and how they can support and affect IT business related solutions
4. Emerging technologies and how business reacts in relation to future strategic planning

7.1.3 Minimum intended module learning outcomes

On successful completion of this module the learner will be able to:

1. Critically evaluate the relationship between information technology infrastructure and organisational competitive advantage
(Linked to PLO 1, PLO 2)
2. Critically analyse and select open source and proprietary software with a view to developing IT solutions for business and business-related IT problems
(Linked to PLO 3, PLO 4)
3. Utilise tools of strategic business analysis to evaluate the current macro and micro business environment with a view to formulating future action plans
(Linked to PLO 5)
4. Research emerging technologies and critically evaluate their impact on business and business information systems in general
(Linked to PLO 8)
5. Understand the relationship between data gathering / utilisation and business intelligence and its impact on industry policy
(Linked to PLO 5, PLO 6)

7.1.4 Rationale for inclusion of the module in the programme and its contribution to the overall IPLOs

This module deals with the application of software in a business context; this very application leads to new software development. This module allows a learner to appreciate the effects of previously developed software on business while giving insight into the need for continuing cross-domain software development.

7.1.5 Information provided to learners about the module

This module specification is replicated in the programme handbook and made available on Moodle.

This information is further supplemented by information given to learners at induction as an overview of the module.

7.1.6 Module content, organisation and structure

Syllabus Content

Competitive Advantage

- The role of data in business
- Data analytics and maintaining a successful competitive advantage
- Pros and cons of Open source versus proprietary software.

Strategic Analysis

- Macro and Micro business environments
- Strategic use of data analytics

Emerging Technologies

- A.I. integration
- Machine learning solutions and options
- Business intelligence and predictive analysis
- Impact of Big Data on business digital transformation strategies

Structure of the Business Enterprise

- Globalisation – Opportunities and Challenges (data protection)
- Role of the data analyst

Professionalism and ethical responsibility

- Personal and professional ethics
- Professional codes of practice
- Professional societies (ICS, ACM, IEEE)

7.1.7 Module teaching and learning (including formative assessment) strategy

To provide the learner with a strong foundation in the core topics covered during the lectures, practical sessions will reinforce lecture content and provide supervised time to complete some assessment tasks. Sessions will be interactive, with instructor-led example exercises highlighting important topics discussed in lectures.

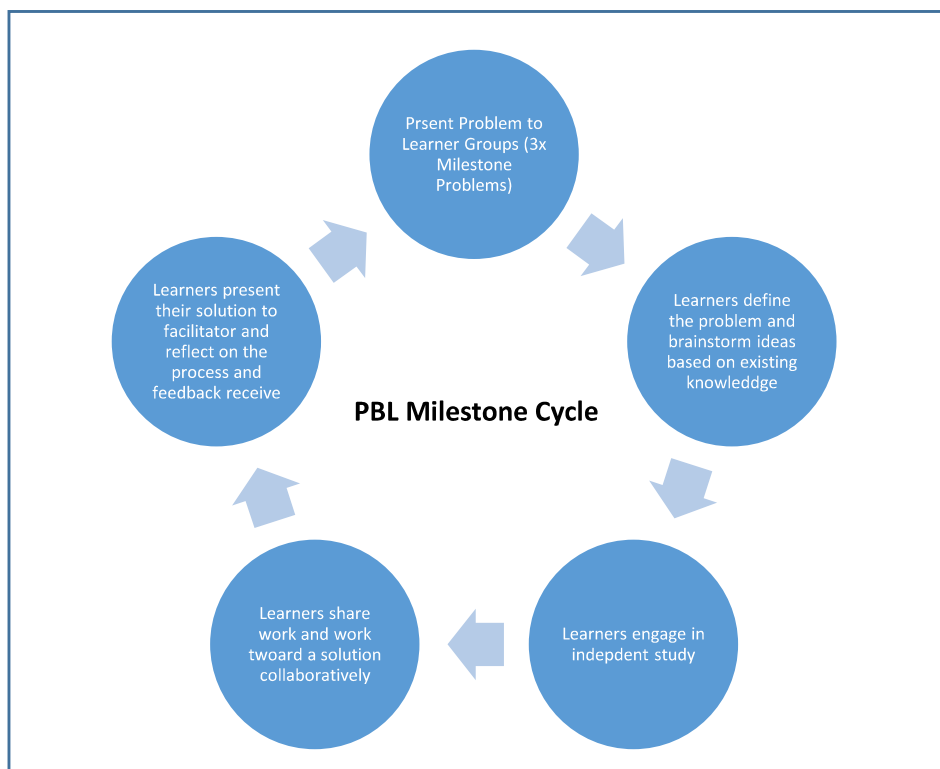
To provide formative assessment for this module the learner will:

- Be provided an opportunity at the beginning of each week to engage in group discussion on the material covered the previous week, thereby allowing reflection and ensuring their competency
- Complete student-suggested tasks in a peer learning environment to encourage collaboration and allow learners to self-evaluate their current knowledge while gaining new knowledge and insights (this strategy links directly to PLO 7, PLO 8)
- Join additional discussions covering any lab-based exercises which have been provided to the learner.

In addition to the above formative processes that exist in all modules, this Capstone module floats across all semesters and is underpinned through a group-based Problem Based Learning (PBL) approach. The PBL approach fosters group collaboration and reflection in which pointed discusses are held in scheduled weekly sessions to allow for

formative feedback throughout the programme. PBL requires group formation with roles and responsibilities and these roles will be shared, swapped and developed as the learners explore the problem under investigation, this ensures all learners have the opportunity to act out a number of roles typically found within industry formed teams, such as team player and team leader and project manager. As this project spans all semesters a structured PBL approach will be used, this will require groups to work through the problem through a number of problem stages with each stage output informing the next, similar to the process of action research. This process allows learner groups investigate and develop solutions that will vary across the cohort furthering the learning of the class as a whole. The final stage of the PBL project will require the learner groups to present their solution in the form of a Poster Presentation in which the learner groups will talk to in front of their peers and assessors. Industry representatives will also be invited to witness these presentations informally at the end of the module furthering the formative feedback nature of the programme. The programme PBL milestone cycle is presented below:

PBL Milestone Cycle



Online and on campus learning activities according to learning type.

Learning Type	Online activities	On campus activities
Knowledge Acquisition	• Pre-recorded presentations / demonstrations • Multi-media text-based materials • Videos • Guest speakers • Open ed resources	• Face to face lecturers • Practical Demonstrations

Collaboration	• Group projects • Discussion forums • Virtual classroom peer learning • Team virtual lab activities • Group presentations	• Group projects • Group presentations	
Discussion	• Discussion forums (synchronous and asynchronous) • Zoom breakout room discussions • Online tutorials • Project supervision • Webinars (industry experts) • Reflective activities	• Class discussion • Project supervision • Face to face lab / practical activities	
Investigation	• Open ed resources • Project research • Information and data sourcing, analysis and evaluation • Flipped Learning	• Text / data sourcing and analysis	
Practice	• Case studies • Analysis of data sets • Presentations • Online quizzes / MCQs	• Practical workshops • Group work	
Production	• E-portfolio • Reflective journal • Assessment outputs • Case studies • Github records	• Case studies • Final Project • Student demonstration	

7.1.8 Work-based learning and practice-placement

Not Applicable

7.1.9 E-learning

Collaborative blended learning strategies will be utilised for this module to ensure peer learning can be experienced not only through face-to-face traditional learning but also through online approaches as indicated in the table in 7.1.7. This integrated learning approach ensures learning can be both reflective and collaborative while developing the efficacy of the individual student as the module progresses.

7.1.10 Module physical resource requirements

Physical resource requirements are 1 laptop or PC/workstation per student. Student are required to provide their own laptop to fully engage with the programme. Opensource software is used.

7.1.11 Reading lists and other information resources

Books and eBooks

Betz, F. (2016) *Strategic Thinking: A Comprehensive Guide*. London: Emerald Group Publishing Ltd. Database: eBook Business Collection (EBSCO)

Chevallier, A (2016) *Strategic Thinking in Complex Problem Solving*. New York: Oxford University Press. ISBN: 9780190463908

Grundy, T. (2014) *Demystifying Strategic Thinking: Lessons from Leading CEOs*. London: Kogan Page. Database: ebook Business Collection (EBSCO)

Kerzner, H. (2017) *Project Management: A Systems Approach to Planning, Scheduling and Controlling* 12th edn. London: Wiley ISBN :9781119165354

Sloan, J. (2019) *Learning to Think Strategically* London: Routledge ISBN: 9780367141462

7.1.12 Specifications for module staffing requirements

Academic and Professional: MSc required. However, NFQ Level 8 in Computer Science, Software Development, Software Engineering or equivalent is acceptable in cases where significant industrial experience is evident. Industry experience will be essential for those who do not have a postgraduate award.

Software Development and Associated Experience: A minimum of 1 year experience required, 3 years of experience preferred.

Pedagogical: Teaching experience is desired. Completion of postgraduate CPD/Certificate in Teaching and Learning or similar preferred. Experience in blended learning delivery required. In absence of experience, training will be provided.

7.1.13 Module summative assessment strategy

The principal assessment will be a group project that spans all semesters and integrates knowledge and skills from all modules within the course of study. The assessment will utilise Problem Based Learning. Learners will work in groups to research and address a real-world or industry informed data centric problem.

The groups will be formed through several formation techniques depending on the cohort and context, for example, randomisation, prior knowledge and experience or self-selection. The group size will be typically between 3 to 4 members.

The problem to be investigated will be provided to all learner groups, which will be broken down further into three supplied Problem Milestones. This process ensures learners are working towards deadlines and facilitates guidance as new learning and knowledge is acquired as the curriculum progresses. The structure of the PBL project in phases is expressed in the below:

Phase	Description	Learner Activity
Phase 1	Overarching Problem Expressed, with Problem Milestone 1 presented	Learner groups are formed and groups begin reflecting on the overarching problem and initially express their understanding and concerns. Learner groups work towards the development of a solution to milestone 1.
Phase 2	Problem Milestone 2 presented to learners	Learner groups use the output of Problem Milestone 1 to inform the planning for Problem Milestone 2. Learner groups work towards the development of a solution to milestone 2.
Phase 3	Problem Milestone 3 presented to learners	Learner groups use the output of Problem Milestone 2 to inform the planning for Problem Milestone 3. Learner groups work towards the development of a solution to milestone 3.
Phase 4	Solution Presentation to Problem under investigation	Learner groups summarise their findings from all three milestones and present their proposed solution to address the problem as a whole.

Summary of Assessed Deliverables and Timeline:

Delivery Mode: **Full-time**

Semester & Phase	Deliverables	Description	% Weighting	
			Individual Contribution	Group Contribution
Sem 1, Phase 1	Problem Milestone Plan 1	Learner groups will put forward a project plan with roles and responsibilities to address Problem Milestone 1	n/a	5%
	Individual Weekly Reflection	Each learner will be required to complete a weekly reflection-in-action journal entry	5%	n/a
	Group Reflection Report	The group will be required to complete an end of milestone summary reflection report	n/a	5%
	Problem Milestone 1 Output Report	The group will produce their solution to Problem Milestone 1	5%	5%
Phase 1 Total Marks			10%	15%

Sem 1, Phase 2	Problem Milestone Plan 2	Learner groups will reflect on their Problem Milestone 1 Output report and put forward a project plan with roles and responsibilities to address Problem Milestone 2	n/a	5%
	Individual Weekly Reflection	Each learner will be required to complete a weekly reflection-in-action journal entry	5%	n/a
	Group Reflection Report	The group will be required to complete an end of milestone summary reflection report	n/a	5%
	Problem Milestone 2 Output Report	The group will produce their solution to Problem Milestone 2	5%	5%
Phase 2 Total Marks			10%	15%
Sem 2, Phase 3	Problem Milestone Plan 2	Learner groups will reflect on their Problem Milestone 2 Output report and put forward a project plan with roles and responsibilities to address Problem Milestone 3	n/a	5%
	Individual Weekly Reflection	Each learner will be required to complete a weekly reflection-in-action journal entry	5%	n/a
	Group Reflection Report	The group will be required to complete an end of milestone summary reflection report	n/a	5%
	Problem Milestone 2 Output Report	The group will produce their solution to Problem Milestone 3	5%	5%
Phase 3 Total Marks			10%	15%
Sem 2, Phase 4	Problem Solution Report	Learner groups will produce a summary report to merge their key findings from Problem Milestone 1, 2 and 3.	5%	5%
	Solution Poster	Learner groups will create a Poster Presentation to succulently capture their staged approach to addressing the overarching project problem and proposed solution.	n/a	5%
	Solution Peer Presentation	Learner groups will present their poster to their peers and facilitators in a conference style event.	5%	5%
Phase 4 Total Marks			10%	15%
Total Individual and Group Marks			40%	60%

Delivery Mode: **Part-time**

Semester & Phase	Deliverables	Description	% Weighting	
			Individual Contribution	Group Contribution

Sem 1, Phase 1	Problem Milestone Plan 1	Learner groups will put forward a project plan with roles and responsibilities to address Problem Milestone 1	n/a	5%
	Individual Weekly Reflection	Each learner will be required to complete a weekly reflection-in-action journal entry	5%	n/a
	Group Reflection Report	The group will be required to complete an end of milestone summary reflection report	n/a	5%
	Problem Milestone 1 Output Report	The group will produce their solution to Problem Milestone 1	5%	5%
Phase 1 Total Marks			10%	15%
Sem 2, Phase 2	Problem Milestone Plan 2	Learner groups will reflect on their Problem Milestone 1 Output report and put forward a project plan with roles and responsibilities to address Problem Milestone 2	n/a	5%
	Individual Weekly Reflection	Each learner will be required to complete a weekly reflection-in-action journal entry	5%	n/a
	Group Reflection Report	The group will be required to complete an end of milestone summary reflection report	n/a	5%
	Problem Milestone 2 Output Report	The group will produce their solution to Problem Milestone 2	5%	5%
Phase 2 Total Marks			10%	15%
Sem 3, Phase 3	Problem Milestone Plan 2	Learner groups will reflect on their Problem Milestone 2 Output report and put forward a project plan with roles and responsibilities to address Problem Milestone 3	n/a	5%
	Individual Weekly Reflection	Each learner will be required to complete a weekly reflection-in-action journal entry	5%	n/a
	Group Reflection Report	The group will be required to complete an end of milestone summary reflection report	n/a	5%
	Problem Milestone 2 Output Report	The group will produce their solution to Problem Milestone 3	5%	5%
Phase 3 Total Marks			10%	15%
Sem 3, Phase 4	Problem Solution Report	Learner groups will produce a summary report to merge their key findings from Problem Milestone 1, 2 and 3.	5%	5%
	Solution Poster	Learner groups will create a Poster Presentation to succulently capture their staged approach to addressing the overarching project problem and proposed solution.	n/a	5%
	Solution Peer Presentation	Learner groups will present their poster to their peers and facilitators in a conference style event.	5%	5%

Phase 4 Total Marks		10%	15%
Total Individual and Group Marks		40%	60%

Online and on campus assessment activities according to assessment type.

Assessment	Online Assessment	On Campus Assessment
Knowledge Acquisition	Pre-recorded presentations / demonstrations	Live presentations / demonstrations
Collaboration	- Group projects - Team virtual lab activities - Group pre-recorded presentations / demonstrations	- Team lab activities - Group presentations / demonstrations
Formative	- Project (individual & group) supervision - Reflective activities	Project (individual & group) supervision
Investigation	- Theoretical and practical project research - Problem Based Learning - Case Study Analysis	Problem Based Learning
Practical	- Virtual lab - Analysis of data sets - Online quizzes / MCQs - Technical tasks (individual and group)	- Lab - Analysis of data sets - Technical tasks (individual and group)
Production	- Artefacts - Student code repository records	- Artefacts - Student code repository records

Reassessment

Different but pedagogically equivalent and appropriately weighted repeat assessment tasks may be given for repeat assessments, i.e Case Study Analysis. Learners who fail to achieve the minimum intended learning outcomes associated with a given assessment will receive constructive feedback (in addition to the normal assignment feedback) on the submitted work and the brief of the repeat assessment. In the case of group work, the group may be directed to recover the failed element of a group assignment, and/or individuals within groups may be directed to recover the individually assessment elements of a group project.

Mapping of summative and formative assessment: MIMLO to MIPLO

	Module Learning Outcomes	Programme Learning Outcomes
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	1	2	3	4	5	1	2	3	4	5	6	7	8
Summative Assessment	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

7.1.14 Sample assessment materials

Sample problems to be investigated for the PBL assessment will be generated through dialogue with relevant programme faculty and consultation with our Industry Engagement Forum. Examples will be provided to the panel.